

MIDTERM EXAM
ECON301—ECONOMETRICS I
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Instructions. The exam has a total of 125 points. Read the problems carefully; then decide which ones you want to attempt first. I suggest that you try to answer first the problems which you feel most confident about and then move on to the ones in which you don't feel so confident. **Write your answers neatly and be organized; cross anything that you don't want me to consider. Please number each page you use.** The duration of the exam is 50 minutes plus 30 minutes of submission time.

1. Consider the following bivariate probability distribution whose joint probability density function is

$$f(x, y) = \frac{3}{11}(x^2 + y)1_{[0,2]}(x)1_{[0,1]}(y).$$

- (a) 10 points. Compute the CEF of Y given X
 (b) 5 points. **True or False:** This model satisfies assumption 2 (linearity) from lecture 3.
 (c) 10 points. Compute the BLP of Y given X .
2. Suppose that you have cross-sectional data on a random vector (X, Y) . Suppose that the data and conditional model satisfy the following assumptions:

A1 The data is a random sample of size n in which $(X_1, Y_1), \dots, (X_n, Y_n)$ are independent and identically distributed random vectors.

A2' $E[Y|X] = \delta X$.

A3' $\sum_i X_i^2 > 0$

A4 $\text{Var}(Y|X) = \sigma^2$

- (a) 15 points. The *Least Squares Estimator through the origin* is defined as the solution to $SS(d) = \min_d \sum_{i=1}^n (Y_i - dX_i)^2$. Thus, an estimator for δ is the argument $\hat{\delta}$ that minimizes $SS(d)$. Give a formula for $\hat{\delta}$ by solving the optimization problem.
- (b) 15 points. What is $E(\hat{\delta}|X's)$? What is $\text{Var}(\hat{\delta}|X's)$?
- (c) 15 points. Let $(\hat{\beta}_0, \hat{\beta}_1)$ be the standard least squares estimator as seen in lecture 3. For the model assumptions in this problem, what is $E[\hat{\beta}_1|X's]$? What is $\text{Var}(\hat{\beta}_1|X's)$?
- (d) 5 points. For two *conditionally unbiased* estimators θ and γ , we say that θ is *more efficient* than γ if the conditional variance of θ is smaller than the one from γ . **True or False:** $\hat{\delta}$ is more efficient than $\hat{\beta}_1$.
3. A researcher wants to estimate a simple regression explaining monthly salary (*wage*) in terms of IQ score (*IQ*). She has data on these two variables for a cross-section of 935 individuals. The results from her estimation as well as some sample statistics (remember that bars above variables represent the sample mean) are presented below.

	Estimate	Std. Error
Intercept	116.9916	85.6415
IQ	8.3031	0.8364

$$n = 935, \quad \hat{\sigma} = 384.8, \quad \overline{wage} = 957.9455, \quad \overline{IQ} = 101.2824, \quad \sum_{i=1}^{935} IQ_i^2 = 9802965,$$

$$\widehat{\text{Var}(\hat{\beta}|X's)} = \begin{bmatrix} 7334.47155 & -70.8527594 \\ -70.85276 & 0.6995568 \end{bmatrix}$$

Suppose that *all* assumptions 1 to 5 from the conditional approach presented in lecture 3 hold.

- (a) 5 points. What is the estimated ceteris paribus effect IQ on wage? How do you interpret it?

- (b) *10 points.* Suppose that the researcher wants to test the following hypotheses: H_0 : “The effect of 1 more IQ point is 10\$ more of monthly salary” versus H_a : “The effect of 1 more IQ point is *more than* 10\$ of monthly salary. ” Construct the previous hypothesis test and test it at the 5% and 1% significance levels.
- (c) *10 points.* Construct a 99% confidence interval for the ceteris paribus effect of IQ on wage.
- (d) *5 points.* There is an individual in the cross-section with an IQ of 120. What is its predicted monthly wage?
- (e) *10 points.* Provide a *formula* for $E(\hat{\beta}_0 + 120\hat{\beta}_1|IQ's)$ and $Var(\hat{\beta}_0 + 120\hat{\beta}_1|IQ's)$. Then, with the data in the display above, provide an *estimate* of the previous two moments.
- (f) *10 points.* Use the results from (d) and (e) to propose a method to construct a 95% confidence interval for the predicted monthly wage of an individual with an IQ of 120, and then use your proposed method to construct it.